



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000

WEEK 23

TUTOR GUIDE

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Last reviewed, October 2010.

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RELEVANT PRIOR LEARNING:

MED 1000:

- 1.09 – Clinical anatomy of the nervous system and nerves [L & P]
- 1.16 – Management of pain [L]
- 1.24 – Anatomy of the cranial nerves [L]
- 1.26 – Neurological examination of the lower limb [L & CCS]
- 1.27 – Anatomy of the vertebral canal and spinal cord [L]
- 1.28 – Biochemistry of muscle contraction [L]
- 1.29 - Anatomy of the brain [L & P]
- 1.29 – History and examination of the nervous system [L & CCS]
- 1.30 – Examination of the cranial nerves [L & CCS]

MED 2000

- 2.13 - Mechanism of actions of medications for pain relief [L]
- 2.19 – Applied anatomy of the head, cranial nerves [L]
- 2.19 & 2.20 – Examination of the nervous system [L & CCS]
- 2.20 – Assessment of headache [L]

MED 3000

- 3.10 – Headache in adults - Intracranial lesions [Short case tutorial]
- 3.30 – Headache in children – intracranial lesions and hydrocephalus [short case tutorial]

WEEK 23_SURGERY_NEUROSURGERY		
Yr	Wk	LEARNING OBJECTIVES:
4	23	• Know the presenting symptoms and signs of cervical nerve root entrapment. • List the 3 main categories of causes of spinal cord compression and be able to provide examples. • Demonstrate an understanding of the applied anatomy of the cervical spine and nerves by explaining why it is critical to make an early diagnosis of spinal cord compression. • List the symptoms and signs of spinal cord compression.
4	23	• Know the key clinical features associated with Subarachnoid haemorrhage (SAH) and the main underlying causes. • Outline why all SAH cases should be referred urgently for neurosurgical management. • Differentiate between primary and secondary headaches and outline some of the common causes of secondary headaches. • Briefly list the features in a headache that would raise a “red flag” i.e. indicate underlying serious pathology.
4	23	• Know the symptoms and signs suggestive of raised intracranial pressure. • Outline the effect of radiotherapy on brain tissue. • Describe the differential diagnoses which should be considered in a patient presenting with polyuria and polydipsia. • Know the causes of Diabetes insipidus and its underlying pathophysiology.
4	23	• Know the underlying pathophysiology of myasthenia gravis (MG) and the diagnosis is confirmed. • Outline the treatment approaches used in MG.
4	23	• Know the key features that would suggest a cerebellar lesion. • Outline the main categories of cerebellar syndromes. • Demonstrate an understanding of the applied anatomy of the nervous system by describing the key differences in motor function when differentiating between upper motor (pyramidal), extrapyramidal and cerebellar lesions.
4	23	• Know the main features in the clinical presentation of trigeminal neuralgia and briefly outline the treatment. • Describe key hallmarks/features of a patient’s pain symptoms that would suggest pain neuropathic origin.

CASE ONE

Short case number: 4_23_01

Category: Nervous System

Discipline: Medicine

Setting: General practice

Topic: Spinal nerve root compression_cervical spondylosis

CASE

The staff of a nursing home called you because Edna, 75 years, had been complaining of severe pain on the right side of the neck radiating down to her right shoulder. This had persisted despite analgesia and they also noticed that she was unable to get dressed because she had difficulty moving her right arm.

On examination, she was unable to move her neck because of pain. There was noticeable weakness of flexion of the right elbow together with sensory loss over the radial aspect of the right arm. Hand movements were normal, there was no sensory loss over the hands and the right biceps reflex was absent. The rest of her physical examination was normal.

Plain radiographs, including lateral and oblique views, showed widespread osteophytes of the cervical spine but no space occupying lesions. You make a provisional diagnosis of cervical spondylosis and Edna's symptoms were managed with conservative measures.

QUESTIONS

1. Referring to your knowledge of dermatomes, work out which cervical nerve is implicated with these symptoms. What additional clinical findings would help you to narrow down your diagnosis?
2. What is the natural history of cervical spondylosis? What are the conservative measures mentioned?



Edna was doing well and recovering from her symptoms but several weeks later, the staff asked you to review Edna because she was having difficulty passing urine and they noticed that she was not walking as well as she normally did.

You recognise the potential seriousness of these symptoms and you immediately refer her for a neurosurgical opinion. The MRI confirmed cervical spondylosis with nerve root compression.

3. Outline the three main categories of causes of spinal cord compression. Why is it critical to diagnose this early? What are the key symptoms/signs of spinal cord compression?

Resources:

- **Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 10 Rheumatology & Bone Diseases: Nerve root entrapment p 506. Saunders Elsevier 2009.**
- **Allen CMC, Lueck CJ, Dennis M. Neurological disease: Cervical spondylosis. Chap 26, Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingstone Elsevier 2006. P1241-2.**

ANSWERS

1. Referring to your knowledge of dermatomes (Med1000 Weeks 5, 6 and 12), work out which cervical nerve is implicated with these symptoms. What additional clinical findings would help you to narrow down your diagnosis?

Nerve root entrapment This is caused by an acute cervical disc prolapse or pressure on the root from spondylotic osteophytes narrowing the root canal. **Acute cervical disc prolapse presents with unilateral pain in the neck**, radiating to the interscapular and shoulder regions. This diffuse, aching dural pain is followed by sharp, electric shock-like pain down the arm, in a nerve root distribution, often with pins and needles; numbness, weakness and loss of reflexes (see Table 10.4). **Cervical spondylosis** occurs in the **older patient with postero-lateral osteophytes** compressing the nerve root and causing root pain, commonly at C5/C6 or C6/C7; it is seen on **oblique radiographs of the neck**. An MRI scan shows facet joint OA, root canal narrowing and any associated disc prolapse clearly.

Table 10.4 Cervical nerve root entrapment – symptoms and signs

Nerve root	Sensory changes	Reflex loss	Weakness
C5	Lateral arm	Biceps	Shoulder abduction Elbow flexion
C6	Lateral forearm Thumb and index finger	Biceps Supinator	Elbow flexion Wrist extension
C7	Middle finger	Triceps	Elbow extension
C8	Medial forearm Little and ring fingers	None	Finger flexion
T1	Medial upper arm	None	Finger ab- and adduction

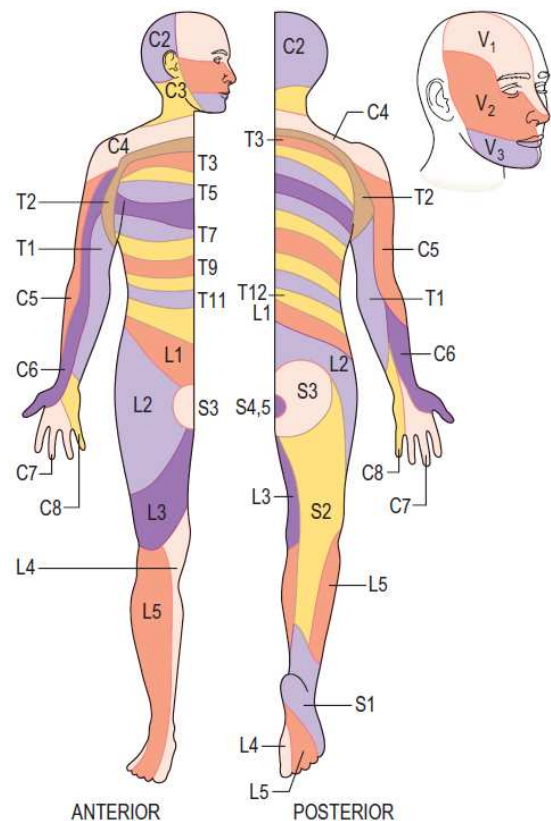


Fig. 21.11 Dermatomes of spinal roots and Vth nerve.

Plain radiographs, including lateral and oblique views, should be obtained to confirm the presence of degenerative changes and to exclude other conditions, including destructive lesions. If surgery is contemplated, MRI is required. In the older patient, CT scans and MRI will show widespread changes; so it is necessary to target the radiographs to the appropriate level of cervical spine under investigation.

Electrophysiological studies rarely add to the clinical examination, but may be necessary if there is doubt about the differential diagnosis between root and peripheral nerve lesions.

2. What is the natural history of cervical spondylosis? What are the conservative measures mentioned?

Conservative treatment results in resolution of symptoms in the great majority of patients.

- Support collar
- Rest
- analgesia
- sedation, as necessary.

Patients should be advised not to carry heavy items. It **usually recovers in 6–12 weeks**. Note that this is in contrast to lumbar spondylosis where compression does not get better with conservative measures. Some patients may require surgery in the form of foraminotomy or disc excision.

3. Outline the three main categories of causes of spinal cord compression. Why is it critical to diagnose this early? What are the key symptoms/signs of spinal cord compression?

Neurosurgical referral is essential if the pain persists or if the neurological signs of weakness or numbness are severe or bilateral. Bilateral root pain is a neurosurgical emergency because a central disc prolapse may compress the cervical spinal cord.

Acute spinal cord compression is one of the most common neurological emergencies encountered in clinical practice and the common causes are listed in [Box 26.114](#). A space-occupying lesion within the spinal canal may damage nerve tissue either directly by pressure or indirectly by interfering with blood supply. **The early stages of damage are reversible but severely damaged neurons do not recover; hence the importance of early diagnosis and treatment.**

The onset of symptoms of spinal cord compression is usually slow (over weeks), but can be acute as a result of trauma or metastases, especially if there is associated arterial occlusion. The symptoms are shown in [Box 26.115](#). The distribution of these signs varies with the level of the lesion, as shown in [Box 26.116](#).

26.114 CAUSES OF SPINAL CORD COMPRESSION

Site	Frequency	Causes
Vertebral	80%	Trauma (extradural) Intervertebral disc prolapse Metastatic carcinoma (e.g. breast, prostate, bronchus) Myeloma Tuberculosis
Meninges (intradural extramedullary)	15%	Tumours (e.g. meningioma, neurofibroma, ependymoma, metastasis, lymphoma, leukaemia) Epidural abscess
Spinal cord (intradural intramedullary)	5%	Tumours (e.g. glioma, ependymoma, metastasis)

Pain and sensory symptoms occur early, while weakness and sphincter dysfunction are usually late manifestations. The signs vary according to the level of the cord compression and the structures involved. Involvement of the roots at the level of the compression may cause dermatomal sensory impairment and corresponding lower motor signs. Interruption of fibres in the spinal cord causes sensory loss and upper motor neuron signs below the level of the lesion, and there is often disturbance of sphincter function.

26.115 SYMPTOMS OF SPINAL CORD COMPRESSION

Pain
Localised over the spine or in a root distribution, which may be aggravated by coughing, sneezing or straining
Sensory
Paraesthesia, numbness or cold sensations, especially in the lower limbs, which spread proximally, often to a level on the trunk
Motor
Weakness, heaviness or stiffness of the limbs, most commonly the legs
Sphincters
Urgency or hesitancy of micturition, leading eventually to urinary retention

26.116 SIGNS OF SPINAL CORD COMPRESSION

Cervical, above C5
- Upper motor neuron signs and sensory loss in all four limbs - Diaphragm weakness (phrenic nerve)
Cervical, C5 to T1
- Lower motor neuron signs and segmental sensory loss in the arms; upper motor neuron signs in the legs - Respiratory (intercostal) muscle weakness
Thoracic cord
Spastic paraplegia with a sensory level on the trunk
Conus medullaris
Lesions at the end of the spinal cord cause sacral loss of sensation and extensor plantar responses
Cauda equina
Spinal cord ends at approximately the T12/L1 spinal level and spinal lesions below this level can only cause lower motor neuron signs by affecting the cauda equina

CASE TWO

Short case number: 4_23_02

Category: Nervous System

Discipline: Medicine

Setting: General practice

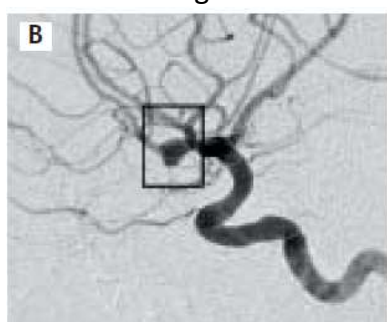
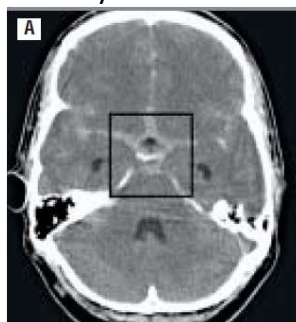
Topic: Acute severe headache_Subarachnoid haemorrhage (SAH)

CASE

Kathy is a 35 year old marketing executive who presents to you with sudden onset of a severe occipital headache. In the past she has had tension headaches which have been quite manageable; but this time the headache is different because she is feeling very nauseous and the headache has been getting worse despite strong analgesics.

QUESTIONS

1. List the key features in the clinical presentation that would make you suspect subarachnoid haemorrhage (SAH). What are the main underlying causes of SAH?
2. Kathy was admitted to the neurosurgical unit and underwent further investigations:



The non-contrast CT scan (A) shows the classic findings of SAH where the sub-arachnoid space lights up. The angiogram (B) showed an anterior communicating artery aneurysm. Why should a suspected SAH be considered a neurosurgical emergency?

3. What are secondary headaches and why is it important to differentiate them from primary headaches?
4. Outline certain features associated with a headache which would raise a “red flag”.

Resources:

- Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 21 Neurological disease: Headache, migraine and facial pain and Subarachnoid haemorrhage (SAH). p 1136-9 and 1134-5. Saunders Elsevier 2009.
- Allen CMC, Lueck CJ, Dennis M. Neurological disease: Headache and facial pain and Subarachnoid haemorrhage. Chap 26, Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingstone Elsevier 2006. P1160-64 and 1210-1211.
- Joubert J. Diagnosing headache. Australian Family Physician August 2005, 34(8):621-5.
- Stark R. How to treat: Severe headache. Australian doctor. August 2005, pp 29-36

ANSWERS

1. List the key features in the clinical presentation that would make you suspect subarachnoid haemorrhage (SAH). What are the main underlying causes of SAH?

Age: Three-quarters of those presenting with SAH are <65 years

Gender: women are more frequently affected than men

Sudden onset: typically presents with a sudden, severe 'thunderclap' headache which lasts for hours or even days

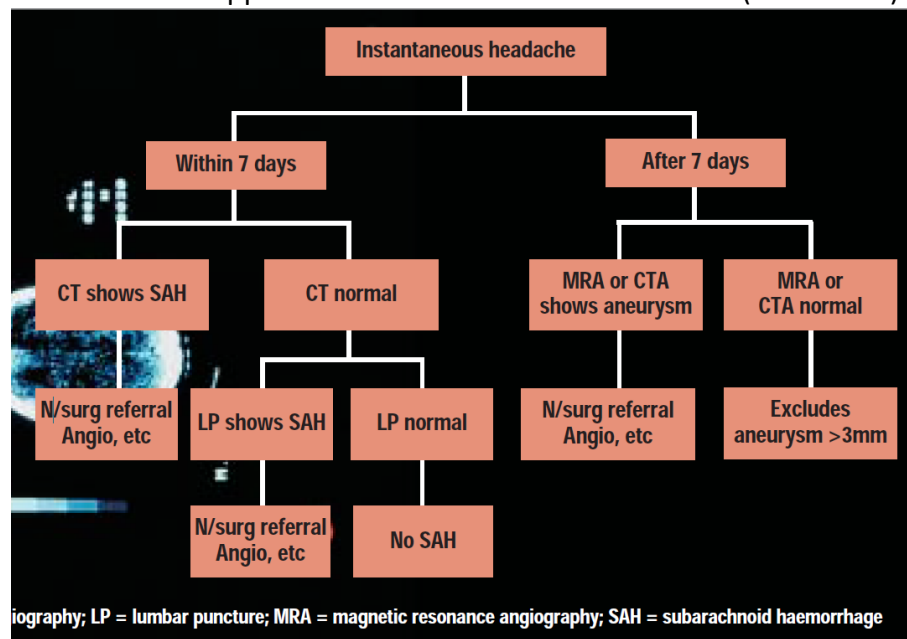
Location of headache: often occipital

Examination findings: patient usually distressed and irritable, with photophobia. There may be neck stiffness but this may take some hours to develop. Focal hemisphere signs (hemiparesis, aphasia etc.) may be present at onset if there is an associated intracerebral haematoma.

Following major SAH there is neck stiffness and a positive Kernig's sign. Papilloedema is sometimes present, with retinal and/or subhyaloid haemorrhage (tracking beneath the retinal hyaloids membrane). Minor bleeds cause few signs, but almost invariably headache.

Immediate treatment of SAH is bed rest and supportive measures. Hypertension should be controlled. Dexamethasone is often prescribed to reduce cerebral oedema; it also is believed to stabilize the blood-brain barrier. Nimodipine, a calcium-channel blocker, reduces mortality.

Recommended approach to an instantaneous headache (Stark 2005):



The underlying causes of SAH are:

Table 21.31	Underlying causes of subarachnoid haemorrhage
Saccular (berry) aneurysms	70%
Arteriovenous malformation (AVM)	10%
No arterial lesion found	15%
Rare associations (<5%)	

Over 70% of SAH are caused by saccular ('berry') aneurysms bulging out from the bifurcations of the cerebral arteries, particularly in the region of the circle of Willis.

2. Why should suspected SAH be considered a neurosurgical emergency?

All SAH cases should be discussed urgently with a neurosurgical centre for urgent investigations including an emergency CT.

Reasons for urgent referral:

Although SAH is rare (incidence 6/100 000) and only 1 patient in 8 with a sudden severe headache has had a SAH, clinical vigilance is necessary to avoid missing the diagnosis or mistaking it for migraine. FAILING to diagnose a subarachnoid haemorrhage is a disaster because most cases arise from ruptured aneurysm and there is a high risk of fatal re-bleeding. If the patient concedes that the headache came on like the snap of their fingers, immediate investigation is required and the choice of tests depends on the time elapsed since the headache began (figure 1). Note that a normal CT scan alone does not exclude SAH, and further investigation with lumbar puncture may be needed. **Confirmed SAH is, of course, a neurosurgical emergency for the following reasons:**

- **Rapid progression:** Headache is usually followed by vomiting and often by coma and death. Survivors may remain comatose or drowsy for hours, days, or longer.
- **High mortality:** Nearly half of SAH cases are either dead or moribund before reaching hospital. The immediate mortality of aneurysmal subarachnoid haemorrhage is about 30%. Survivors have a recurrence, or re-bleed, rate of about 40% in the first 4 weeks and 3% annually thereafter. SAH accounts for some 5% of strokes. Of the remainder, a further 10–20% rebleed and die within several weeks. Failure to diagnose SAH e.g. mistaking SAH for migraine, contributes to this mortality.
- **Therapeutic interventions available:** Insertion of platinum coils into an aneurysm or surgical clipping of the aneurysm neck reduces the risk of both early and late recurrence. Coiling may be associated with fewer perioperative complications and better outcomes.

In the case of suspected SAH where angiography is non-confirmatory, the angiogram should be repeated after 6 weeks. This is often a peri-mesencephalic haemorrhage, a benign venous haemorrhage occurring in patients after coughing.

3. What are secondary headaches and why is it important to differentiate them from primary headaches?

Secondary headaches are an important consideration, not only because of the potential danger inherent in some, but because of their prevalence. A clinical classification of secondary headaches is shown in Table 2. Note in Table 4 in the following page the features that should be considered “red flags”.

Table 2. Secondary headaches	Table 1. Primary and secondary headache disorders
• Headache secondary to disorders of the: – skull (eg. Paget disease, mastoiditis, secondary malignancy) – ears (eg. otitis media or externa) – eyes (eg. glaucoma, strabismus, ocular strain, iritis) – nose and nasal sinuses (eg. acute or chronic sinusitis) – teeth (eg. tooth abscess, malocclusion) – cervical spine (eg. cervical spondylosis) – cranial nerves (eg. herpes zoster, occipital neuralgia) • Headache secondary to: – intracranial vascular disorders (eg. venous sinus thrombosis, ruptured aneurysm, cerebral haemorrhage) – extracranial vascular disorders (eg. carotid artery dissection, cranial arteritis or carotidynia) – disorders of intracranial pressure (raised intracranial pressure or low cerebrospinal fluid pressure, eg. postlumbar puncture leak) – intracranial infections (eg. encephalitis or meningitis) • Headache attributable to: – generalised infection (eg influenza, typhoid or malaria) – medication (eg. vasodilators) – hypertension	Primary episodic headache disorders Migraine Episodic tension type headache Cluster headache and trigeminal autonomic cephalalgias Primary frequent (chronic daily) headache disorders Transformed migraine Chronic tension type headache New daily persistent headache Hemicrania continua Secondary headache disorders

Differential diagnoses: SAH should be considered as a possible diagnosis in any sudden headache and must be differentiated from migraine, which is a common primary episodic headache. Note that “thunderclap headache” is used (confusingly) to describe either SAH or a sudden (benign) headache for which no cause is ever found. Acute bacterial meningitis occasionally causes a very abrupt headache, when a meningeal microabscess ruptures; SAH also occasionally occurs at the onset of acute bacterial meningitis. Cervical arterial dissection can present with a sudden headache.

4. Outline certain features associated with a headache which would raise a “red flag”.

(The next case scenario will provide discussion around a “red flag” headache from a brain tumour.)

Keeping out of trouble: red flags and serious causes of headache — when to think of brain tumour, meningitis, encephalitis, etc

The main red flags are:

- Acute or subacute tempo
- Presence of neurological signs
- Fever
- Neck stiffness
- Headache worst on waking may suggest raised intracranial pressure
- Papilloedema is usually a late sign; its absence is only minimally reassuring

Table 4. Red flag headaches

- New onset, specific setting. New headache in the setting of cancer (metastases), HIV infection (opportunistic infection), postmanipulation or trauma of the neck, or associated with mild head trauma in the elderly (subdural haematoma)
- New headache that is persistent¹
- Focal signs or symptoms (other than the typical visual or sensory aura of migraine)
- Headache with focal neurological signs that precede or outlast the headache (the rare exception is hemiplegic migraine³)
- Headache that is progressive (may suggest a mass lesion)
- Headache of sudden onset (may indicate a bleed either into the subarachnoid space or the cerebral parenchyma)
- Headache with rash (may indicate meningococcal meningitis or Lyme disease)
- Persistent unilateral temple headache in adult life (may indicate cranial arteritis)
- Headache with a raised erythrocyte sedimentation rate (ESR) (may be an indication of cranial arteritis, collagen disease or systemic infection)
- Headache with papilloedema (raises the suspicion of raised intracranial pressure due to a mass lesion or benign intracranial hypertension)
- Nonmigraine headaches in pregnancy or postpartum (cerebral vein thrombosis can occur during or just after pregnancy)
- Headache triggered by cough or straining (may be an indication of either a mass lesion or a subarachnoid bleed¹)
- Headache clearly triggered by changes in posture (may indicate low cerebrospinal fluid [CSF] pressure, for instance due to spontaneous CSF leak)
- Headache associated with pressing visual disturbances (may indicate conditions such as glaucoma or optic neuritis)
- Headaches that have primary characteristics, but with unusual features

CASE THREE

Short case number: 4_23_03

Category: Nervous System

Reproductive & Endocrine Systems

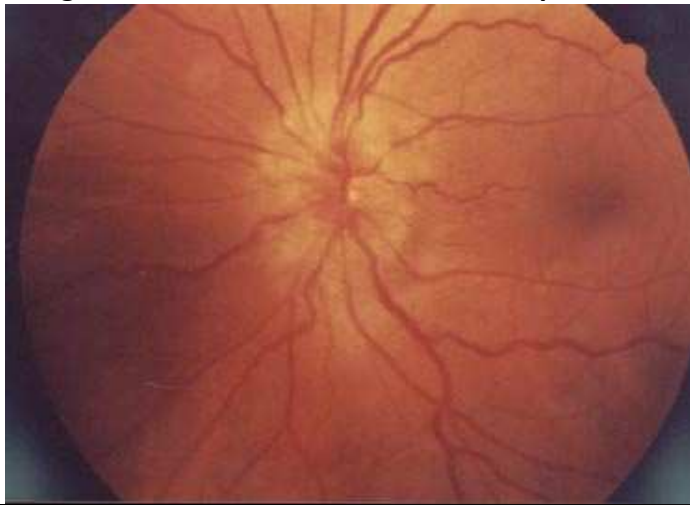
Discipline: Medicine

Setting: General practice

Topic: Brain tumour_diabetes insipidus

CASE

Ben Anderson, a 36 year old dentist, presented to your general practice because he had been having mild headaches on and off for the past few days. However on the morning of the



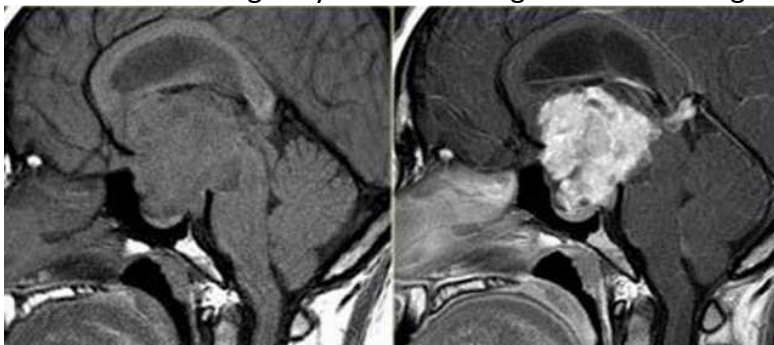
consultation he had been woken by a severe headache. He also developed nausea and vomiting and was convinced he had food poisoning.

On examination he was afebrile, no rashes, PR 75/min, BP 130/70 mm Hg. He was agitated and CNS examination was normal. There was no neck stiffness or Kernig's sign but on fundoscopic examination you noted bilateral papilloedema.

QUESTIONS

1. In a patient presenting with a headache that is worst in the morning and associated with nausea and vomiting, what is an important diagnosis that needs to be excluded?

2. You refer Ben urgently to a neurosurgeon and radiological investigations reveal a space



-occupying lesion within the sella turcica with suprasellar extension. The biopsy confirmed a glioma. Because of the location of the lesion meant that further surgery would be incomplete, the decision was made to treat the patient with radiotherapy

- for 6 weeks. What are the effects of radiotherapy on the brain?
3. Six weeks after radiotherapy, the patient developed symptoms of polyuria and polydipsia. What are the other causes of polyuria and polydipsia? What are the causes of DI and how is it diagnosed?

Resources:

- Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 18 Endocrine diseases p 1017-8-544 and Chapter 21 Neurological disease, p 1160-62. Saunders Elsevier 2009.
- Strachan MWJ and Walker BR. Endocrine disease Chap 20, Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingstone Elsevier 2006, p796-9.
- Hovey E, Koh E-S, Jeffree L, Cole A. How to treat: Malignant glioma in adults, Australian Doctor, March 2010, pp33-40.
- National Guidelines for the Management of Adult Gliomas: Astrocytomas and Oligodendrogliomas:
http://www.cancer.org.au/File/HealthProfessionals/Clinical%20Guidelines/Clinical_Practice_Guidelines-Adult_Gliomas-AUG09.pdf

ANSWERS

1. In a patient presenting with a headache that is worst in the morning and associated with nausea and vomiting, what is an important diagnosis that needs to be excluded?

Raised intracranial pressure (headache, vomiting and papilloedema), a relatively unusual presentation of a mass lesion in the brain, often imply hydrocephalus – obstruction to CSF pathways. Typically this is produced early by posterior fossa masses that obstruct the aqueduct and IVth ventricle but only later with lesions above the tentorium.

- Shift of the intracranial contents produces features that coexist with the direct effects of any expanding mass:
- Distortion of the upper brainstem, as midline structures are displaced either caudally or laterally by a hemisphere mass. This causes impairment of consciousness.
- Compression of the medulla, by herniation of the cerebellar tonsils caudally through the foramen magnum, an example of coning. This causes impaired consciousness, respiratory depression, bradycardia, decerebrate posturing and death.

What not to miss

- New-onset or recurrent headache uncharacteristic for that patient is an indication for neuro-imaging with contrast CT brain scan.
- Patients presenting with a first seizure should have a contrast MRI scan.
- Patients with signs or symptoms suggesting raised intracranial pressure or focal neurological signs should be referred immediately to a neurosurgeon.
- However suggestive of specific histology the imaging may be, it is vital that patients be offered surgery for tissue confirmation of their diagnosis.

Indications for urgent referral to a neurosurgical service

Patients with brain lesion on imaging and any one of the following:

- Headache, especially if severe, persistent or occurring in morning
- Vomiting
- Reduced conscious state, drowsiness
- Papilloedema
- Imaging evidence of midline shift
- Imaging evidence of hydrocephalus

PATIENTS with glioma may present with a variety of symptoms, the most common of which is headaches. **Headaches suggestive of increased intracranial pressure** are most severe in the morning, may wake the patient from sleep and may be associated with nausea and vomiting.

THE diagnosis of malignant glioma is suggested by MRI or CT scan showing a mass lesion within the brain parenchyma, expanding and displacing neural structures.

2. You refer Ben urgently to a neurosurgeon and imaging reveals a space-occupying lesion within the sella with suprasellar extension. The biopsy confirmed a low-grade glioma. Because of the location of the lesion meant that further surgery would be incomplete, the decision was made to treat the patient with radiotherapy for 6 weeks. What are the effects of radiotherapy on the brain?

The standard therapy for newly diagnosed malignant gliomas involves surgery, radiotherapy and chemotherapy. Patients with gliomas tend to suffer from a variety of medical problems as a result of the primary disease or its treatments e.g. seizure control, venous thromboembolism, skin reactions, cognitive deficits and side effects of corticosteroids.

Side effects	Radiotherapy	Radiotherapy to the brain can be associated with white matter changes, cognitive deficits and radiation necrosis. The adverse effects of radiotherapy in this case are difficult to isolate from the direct effects of a tumour in varying locations, and confounded by other factors such as anticonvulsant medications and epilepsy.
Acute (days–weeks)	• Fatigue and hypersomnolence • Mild to moderate scalp erythema • Patchy alopecia • Mild to moderate headache	
Sub-acute (weeks–months)	• Hypersomnolence syndrome	
Late (months–years)	• Radiation necrosis (<5%) • Cognitive dysfunction (low risk) • Endocrine dysfunction	

3. Six weeks after radiotherapy, the patient developed symptoms of polyuria and polydipsia. What are the other causes of polyuria and polydipsia? What are the causes of DI and how is it diagnosed?

Other causes of polyuria and polydipsia

Diabetes mellitus, hypokalaemia and hypercalcaemia should be excluded. In the case of diabetes mellitus the cause is an osmotic diuresis secondary to glycosuria which leads to dehydration and an increased perception of thirst owing to hypertonicity of the extracellular fluid.

Primary polydipsia: This is a relatively common cause of thirst and polyuria. It is a disturbance characterized by the excessive intake of water. Plasma sodium and osmolality fall as a result and the urine produced is appropriately dilute. Vasopressin levels become virtually undetectable. Prolonged primary polydipsia may lead to the phenomenon of 'renal medullary washout', with a fall in the concentrating ability of the kidney. Characteristically the diagnosis is made by a water deprivation test. A low plasma osmolality is usual at the start of the test, and since vasopressin secretion and action can be stimulated, the patient's urine becomes concentrated (albeit 'maximum' concentrating ability may be impaired); the initially low urine osmolality gradually increases with the duration of the water deprivation.

Diabetes insipidus (DI)

Deficiency of vasopressin or insensitivity to its action leads to polyuria, nocturia and compensatory polydipsia. Daily urine output may reach as much as 10–15 L, leading to dehydration that may be very severe if the thirst mechanisms or consciousness are impaired or the patient is denied fluid. The most common is hypothalamic–pituitary surgery, following which transient DI is common, frequently remitting after a few days or weeks. DI may be masked by simultaneous cortisol deficiency – cortisol replacement allows a water diuresis and DI then becomes apparent.

Table 18.32 Causes of diabetes insipidus**Cranial diabetes insipidus**

Familial (e.g. DIDMOAD)

Idiopathic (often autoimmune)

Tumours

Craniopharyngioma

Hypothalamic tumour, e.g. glioma, germinoma

Metastases, especially breast

Lymphoma/leukaemia

Pituitary with suprasellar extension (rare)

Infections

Tuberculosis

Meningitis

Cerebral abscess

Infiltrations

Sarcoidosis

Langerhans' cell histiocytosis

Inflammatory

Hypophysitis

Post-surgical

Transfrontal

Trans-sphenoidal

Post-radiotherapy (cranial)**Vascular**

Haemorrhage/thrombosis

Sheehan's syndrome

Aneurysm

Trauma (e.g. head injury)**Nephrogenic diabetes insipidus**

Familial (e.g. vasopressin receptor gene, aquaporin-2 gene defect)

Idiopathic

Renal disease (e.g. renal tubular acidosis)

Hypokalaemia

Hypercalcaemia

Drugs (e.g. lithium, demeclocycline, glibenclamide)

Sickle cell disease

Box 18.10 Water-deprivation test**Indication**

Diagnosis or exclusion of diabetes insipidus

Procedure

Fasting and no fluids from 0730 h (or overnight if only mild DI is expected and polyuria is only modest)

Monitor serum and urine osmolality, urine volume and weight hourly for up to 8 hours

Abandon fluid deprivation if weight loss >3% occurs

If serum osmolality >300 mOsm/kg and/or urine osmolality <600 mOsm/kg give desmopressin 2 µg i.m. at end of test. Allow free fluid but measure urine osmolality for 2–4 hours

Interpretation*Normal response.* Serum osmolality remains within normal range (275–295 mOsm/kg). Urine osmolality rises to >600 mOsm/kg.*Diabetes insipidus (DI).* Serum osmolality rises above normal without adequate concentration of urine osmolality (i.e. serum osmolality >300 mOsm/kg; urine osmolality <600 mOsm/kg).**Diagnosis:**

- High or high-normal plasma osmolality with low urine osmolality (in primary polydipsia plasma osmolality tends to be low).
- Resultant high or high-normal plasma sodium.
- High 24-h urine volumes (less than 2 L excludes the need for further investigation).
- Failure of urinary concentration with fluid deprivation.
- Restoration of urinary concentration with vasopressin or an analogue.

Treatment:

- Fluid balance control with daily monitoring of input/output. The patient should be thereafter undergo regular electrolyte and plasma osmolality monitoring.
- The synthetic vasopressin analogue desmopressin is the treatment of choice. It is most reliably given intranasally as a spray 10–40 µg once or twice daily, but can also be given orally as 100–200 µg three times daily or intramuscularly 2–4 µg daily. Response is variable and must be monitored carefully.
- Where there is a reversible underlying cause (e.g. a hypothalamic tumour) this should be investigated and treated.

CASE FOUR

Short case number: 4_23_04

Category: Nervous System

Discipline: Medicine

Setting: General practice

Topic: Disease of the neuromuscular junction and muscles

CASE

Helen is a 35 year old librarian who was referred by her optometrist to see you because she started noticing problems with her vision over the last few months. She noticed that



sometimes at the end of the day, when she is very tired, she sees double. The optometrist had noted that her eyelids appeared to be drooping on the day she attended.

On current examination there were no motor or sensory abnormalities and her CNS examination was normal.

QUESTIONS

1. You suspect a diagnosis of myasthenia gravis. What is the underlying pathophysiology of myasthenia gravis (MG)? And how is the diagnosis confirmed?
2. Helen asks about her condition and whether there is any treatment for this condition. What will you tell her?

Resources:

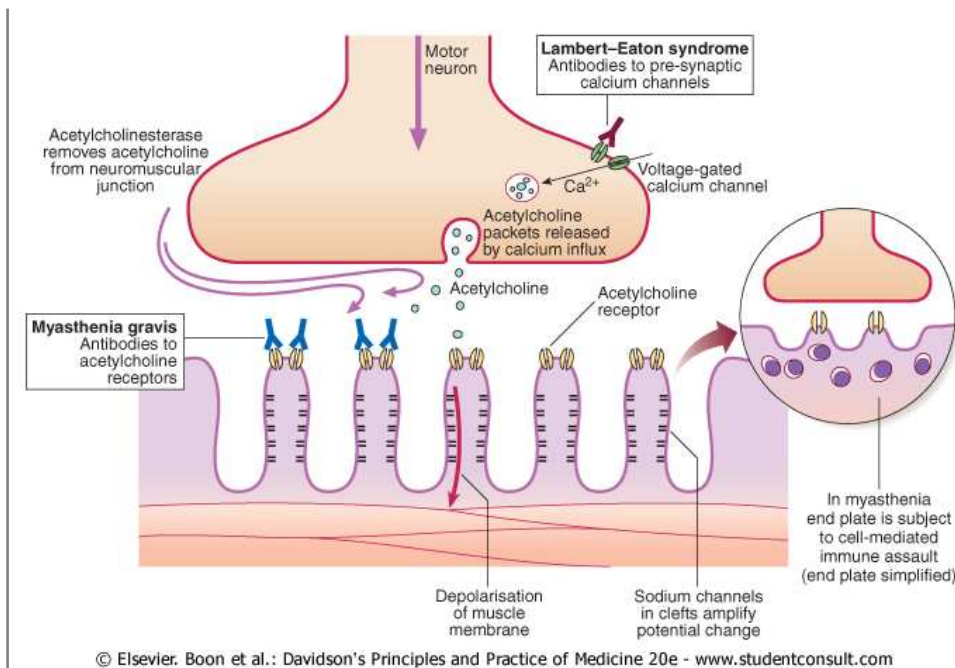
- Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 21 Neurological disease, p 1180-81. Saunders Elsevier 2009.
- Allen CMC, Lueck CJ, Dennis M. Neurological diseases, Chap 26, Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingston Elsevier 2006. P1252-4.

ANSWERS

1. You suspect a diagnosis of myasthenia gravis. What is the underlying pathophysiology of myasthenia gravis (MG)? And how is the diagnosis confirmed?

The disease is most commonly caused by autoantibodies to acetylcholine receptors in the post-junctional membrane of the neuromuscular junction. These antibodies block neuromuscular transmission and initiate a complement-mediated inflammatory response which reduces the number of acetylcholine receptors and damages the end plate.

Nothing is known about factors which trigger the disease itself, but penicillamine can cause an antibody-mediated myasthenic syndrome which may persist even after drug withdrawal. Some drugs, especially aminoglycosides and ciprofloxacin, may exacerbate the neuromuscular blockade and should be avoided in patients with myasthenia.



The diagnosis of MG is confirmed by:

- Anti-acetylcholine receptor antibody (AChRA) is found in over 80% of cases, though less frequently in purely ocular myasthenia (50%).
- Anti-MuSK antibodies are found especially in AChRA-negative patients with prominent bulbar involvement.
- Positive anti-skeletal muscle antibodies suggest the presence of thymoma, but all patients should have a thoracic CT to exclude this condition, which may not be visible on plain X-ray examination. Thymomas must be actively sought as they are potentially malignant, in which case they should be surgically excised.
- EMG with repetitive stimulation may show the characteristic decremental response
- iV injection of the short-acting anticholinesterase, edrophonium bromide (the Tensilon test) leading to improvement in muscle power within 30 seconds, and usually persisting for 2-3 minutes.
- Screening for other autoimmune disorders, particularly thyroid disease, is important.
- Screening for neuromuscular disorders.

2. Helen asks about her condition and whether there is any treatment for this condition. What will you tell her?

The disease usually presents between the ages of 15- 50 years, with women affected more often than men in the younger age groups. It tends to run a relapsing and remitting course, especially during the early years. Most cases have a protracted, lifelong course.

The first symptoms are usually intermittent **ptosis or diplopia**, but weakness of chewing, swallowing, speaking or limb movement also occurs.

The cardinal symptom is abnormal fatigable weakness of the muscles (which is different from a sensation of muscle fatigue); movement is initially strong but rapidly weakens. **Worsening of symptoms towards the end of the day or following exercise is characteristic.**

Any limb muscle may be affected, most commonly those of the shoulder girdle; the patient is unable to undertake tasks above shoulder level, such as combing the hair, without frequent rests. Respiratory muscles may be involved, and respiratory failure is a not uncommon cause of death. Aspiration may occur if the cough is ineffectual. Sudden weakness from a cholinergic or myasthenic crisis (see below) may require ventilatory support.

The principles of treatment are:

- **to maximise the activity of acetylcholine at remaining receptors in the neuromuscular junctions**
- **to limit or abolish the immunological attack on motor end plates.**

Anticholinesterases: The duration of action of acetylcholine is greatly prolonged by inhibiting its hydrolysing enzyme, acetylcholinesterase. The most commonly used anticholinesterase drug is pyridostigmine. Muscarinic side-effects, including diarrhoea and colic, may be controlled by propantheline. Over-dosage of anticholinesterase drugs may cause a cholinergic crisis due to depolarisation block of motor end plates, with muscle fasciculation, paralysis, pallor, sweating, excessive salivation and small pupils. This may be distinguished from severe weakness due to exacerbation of myasthenia (myasthenic crisis) by the clinical features and, if necessary, by the injection of a small dose of edrophonium.

The immunological treatment of myasthenia is outlined. Thymectomy in the early stages of the disease leads to a much better overall prognosis, whether a thymoma is present or not.

IMMUNOLOGICAL TREATMENT OF MYASTHENIA

Thymectomy

- Should be performed as soon as feasible in any antibody-positive patient under 45 years with symptoms not confined to extraocular muscles, unless the disease has been established for more than 7 years.
- Necessary in all patients with confirmed thymic tumours.

Plasma exchange

- Removing antibody from the blood may produce marked improvement but, as this is usually brief, such therapy is normally reserved for myasthenic crisis or for pre-operative preparation

Intravenous immunoglobulin

- An alternative to plasma exchange in the treatment of severe myasthenia

- Some patients respond better to plasma exchange and others to iVlg

Corticosteroid treatment

- Improvement is commonly preceded by marked exacerbation of myasthenic symptoms and treatment should be initiated in hospital, if high doses are considered.
- It is usually necessary to continue treatment for months or years, often resulting in adverse effects

Other immunosuppressant treatment

- Azathioprine, mycophenolate and other immunosuppressants may be of value in reducing the dosage of steroids necessary and may allow steroids to be withdrawn
- The effect of treatment on clinical disease is often delayed for several months

Prognosis is variable. Remissions sometimes occur spontaneously. **When myasthenia is confined to the eye muscles, the prognosis is excellent and disability slight. Young female patients with generalised disease have high remission rates after thymectomy**, whilst older patients are less likely to have a remission despite treatment. Rapid progression of the disease more than 5 years after its onset is uncommon.

CASE FIVE

Short case number: 4_23_05

Category: Nervous System

Discipline: Medicine

Setting: General practice

Topic: Cerebellar disorders [SDL]

CASE

George Casey, a 61 year old lawyer, presented with a history of unsteadiness on walking to work over the past few weeks. This has become more pronounced in the past two weeks and he noticed that he had a tendency to veer to the right. He also had a mild headache over the last couple of days.

On examination, you confirm the tendency to drift to the right when you asked him to walk towards you, and also detected dysdiadochokinesis on the right.

QUESTIONS

1. Describe the key features that would suggest a cerebellar lesion under the following headings: posture & gait, tremor & ataxia, nystagmus, dysarthria.
2. What are the main categories of causes of cerebellar syndromes?
3. When examining a patient for their motor function, how do cerebellar lesions differ in their presentations from Upper motor (pyramidal) and Extraparaxidal lesions?

Resources:

- Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 21 Neurological disease: Cerebellar lesions p 1114-5. Saunders Elsevier 2009.
- Allen CMC, Lueck CJ, Dennis M. Neurological disease. Chap 26, Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingstone Elsevier 2006. P1179.

ANSWERS

1. Describe the key features that would suggest a cerebellar lesion under the following headings: posture & gait, tremor & ataxia, nystagmus, dysarthria.

A lesion in a cerebellar hemisphere causes lack of coordination on the same side of the body. The initial part of movement is normal but as the target is approached, the accuracy of the movement deteriorates, producing an 'intention tremor'. The distances of targets are misjudged (dysmetria), resulting in 'past-pointing'. The ability to produce rapid, accurate, regularly alternating movements is impaired (dysdiadochokinesis).

A lesion within one cerebellar lobe (e.g. tumour or infarction) causes disruption of the normal sequence of movements (dyssynergia) on the same side.

Posture and gait. The outstretched arm is held still in the early stages of a cerebellar lobar lesion (cf. the drift of pyramidal lesions) but there is rebound upward overshoot when the limb is pressed downwards and released. Gait becomes broad and ataxic, faltering towards the side of the lesion.

Tremor and ataxia. Movement is imprecise in direction, force and distance (dysmetria). Rapid alternating movements (tapping, clapping or rotary hand movements) become disorganized (dysdiadochokinesis). Intention tremor (action tremor with past-pointing) is seen, but speed of fine movement is preserved, compared to extrapyramidal and pyramidal lesions. Midline cerebellar vermis lesions have dramatic effects on trunk and axial muscles. There is difficulty standing and sitting unsupported (truncal ataxia), with a rolling, broad, ataxic gait.

Nystagmus. Coarse horizontal nystagmus develops with a lateral cerebellar lobe lesion. The fast component is always towards the side of the lesion.

Dysarthria. Halting, jerking speech develops – scanning speech.

Other signs. Titubation – rhythmic head tremor as either forward and back (yes–yes) movements or rotary (no–no) movements – can occur, mainly when cerebellar connections are involved.

Hypotonia (floppy limbs) and depression of reflexes (with slow, pendular reflexes) are also sometimes seen.

2. What are the main categories of causes of cerebellar syndromes?

Table 21.15	Principal causes of cerebellar syndromes
Tumours	Haemangioblastoma Medulloblastoma Secondary neoplasm Compression by acoustic neuroma
Vascular	Haemorrhage Infarction Arteriovenous malformation
Infection	Abscess HIV Prion diseases
Developmental	Arnold–Chiari malformation Basilar invagination Cerebral palsy
Toxic and metabolic	Antiepileptic drugs Chronic alcohol abuse Carbon monoxide poisoning Lead poisoning Solvent abuse
Inherited	Friedreich’s and other spino-cerebellar ataxias Ataxia telangiectasia Essential tremor
Miscellaneous	Multiple sclerosis Hydrocephalus Postinfective (childhood) Hypothyroidism Paraneoplastic syndromes High altitude cerebral oedema

3. When examining a patient for their motor function, how do cerebellar lesions differ in their presentations from Upper motor (pyramidal) and Extrapyrmidal lesions?

Davidson’s 26.33 PHYSICAL SIGNS IN DIFFERENT TYPES OF MOTOR DEFICIT				
Clinical sign	Upper motor (pyramidal) lesion	Extrapyrmidal lesion	Cerebellar lesion	Functional
Power	Weak Upper limbs: extensors weaker Lower limbs: flexors weaker	No weakness	No weakness	Give-way weakness
Wasting	None	None	None	None
Tone	Spastic increase (after interval)	Rigidity (cogwheel)	Normal/reduced	Normal
Reflexes	Increased	Normal	Normal	Normal
Plantar response	Extensor	Flexor	Flexor	Normal
Coordination	Reduced by weakness	Normal (but slowed)	Impaired	Normal (may be laborious)

CASE SIX

Short case number: 4_23_06

Category: Nervous System

Discipline: Medicine

Setting: General practice

Topic: Trigeminal neuralgia [SDL]

CASE

Steve Brody, a 50 year old sales executive presents with severe spasms of pain, each time lasting only seconds, over the left side of his face at the angle of the jaw since yesterday evening. Fortunately he was able to have a good night's sleep but this morning he noticed that when he was shaving, it brought on another spasm of pain.

QUESTIONS

1. What is trigeminal neuralgia and what is the underlying pathophysiology? Outline the treatment.
2. What is neuropathic pain? Describe the hallmarks/features of a patient's pain symptoms that would suggest neuropathic pain.

Resources:

- Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 21 Neurological disease: Trigeminal neuralgia p 1106-7. Saunders Elsevier 2009.
- Allen CMC, Lueck CJ, Dennis M. Neurological disease. Chap 26, Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingston Elsevier 2006. P1164.

1. What is trigeminal neuralgia and what is the underlying pathophysiology? Outline the treatment.

Trigeminal neuralgia (tic douloureux) develops most commonly in the over 50 years age group and is almost always unilateral. Compression, most commonly from aberrant branches of the superior cerebellar artery in the cerebellopontine angle, causes demyelination. It is thought that this sends ephaptic (short circuited) signals to cause pain.

Paroxysms of knife-like or electric shock-like pain, lasting seconds, occur in the distribution of the Vth nerve. Pain tends to commence in the mandibular division (V3) and spreads upwards to maxillary (V2) and ophthalmic divisions (V1). Spasms occur many times a day, but rarely in sleep. Each paroxysm is stereotyped, brought on by stimulation of one or more trigger zones in the face. Washing, shaving, a cold wind or eating are examples of trivial stimuli that provoke pain. The face may be screwed up in agony (hence tic). There are no signs of Vth nerve dysfunction; the corneal reflex is preserved.

Natural history: Spontaneous remissions last months or years before recurrence.

Treatment with Carbamazepine 600–1200 mg daily reduces severity of attacks in the majority. Phenytoin, gabapentin and clonazepam are used, but are less effective. If drugs fail, surgery (radiofrequency extirpation of the ganglion, neurovascular decompression or sectioning of the sensory root) is useful. Alcohol injection into the trigeminal ganglion or peripheral Vth nerve branches can also be carried out.

Differential diagnoses:

Post-herpetic neuralgia following herpes zoster;

Secondary trigeminal neuralgia can occur in MS, with tumours of the Vth nerve (e.g. neuroma) and with cerebellopontine angle lesions. There are often physical signs – initially a depressed corneal reflex, progressing to trigeminal sensory loss.

2. What is neuropathic pain? Describe the hallmarks/features of a patient's pain symptoms that would suggest neuropathic pain.

Neuropathic pain refers to pain arising from pathological states within the nerve fibres themselves. Examples of 'pure' neuropathic pain include post herpetic neuralgia, trigeminal neuralgia, the pain of multiple sclerosis, HIV neuropathy, complex regional pain syndrome and Spinal cord injury. No gold standard exists for the diagnosis of neuropathic pain, but **certain features are commonly found among neuropathic pains** so designated by neurologists or pain physicians. These features include pain descriptors of burning pain, hot pain, shooting or electric shock pain, paraesthesia, anaesthesia (numbness) and touch-evoked pain.

Neuropathic pain can be suspected in the presence of an abnormal sensory experience on clinical examination. Examples of this include:

- Brush allodynia (pain experienced on lightly brushing the hair or skin in the painful area).
- Punctuate mechanical hyperalgesia (significant pain experienced on indenting the skin with a blunt focal point – the 'paperclip test').
- A positive pinch-rolling test (grasping a skinfold between finger and thumb outside the painful area and rolling the skinfold between the fingers into the painful area and eliciting a pain response).